

Eliza Zilliax Blodget

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Education

University of Chicago, Bachelor of Arts in Physics with Honors June 2024

Honors thesis: *Perturbative Minimax Entropy Models for Synchrony in Large Neuronal Populations*

Thesis advisor: Stephanie Palmer

Graduate coursework: Soft Condensed Matter Physics; Statistical Mechanics; String Theory I-II;

Theoretical Neuroscience: Statistics and Information Theory; Quantum Field Theory I-II-III

Research

Research Analyst, Palmer Theory Group, University of Chicago Aug 2024 – Present

Dept. of Organismal Biology & Anatomy and Dept. of Physics

PI: *Stephanie Palmer*

Used topologically restricted statistical physics models to study efficient, robust encoding of scene identity in naturalistic environments, analyzing data from neurons in the salamander retina. Examined scaling of scene discrimination speed with cell population size and model relative entropy.

Additionally, began work on boundary dynamics in finite hydrodynamic flocks.

Undergraduate Research Assistant, Palmer Theory Group, UChicago Sep 2023 – Jun 2024

Dept. of Physics

PI: *Stephanie Palmer*

Studied the Ising model and critical phenomena on complex networks. Constructed maximum entropy models of neuronal activity constrained to tree-like networks, and used these models to analyze data from large populations of neurons in the retina and hippocampus.

Publications

Topologically constrained statistical physics models capture functional aspects of a sensory code Dec 2025

Eliza M. Z. Blodget, Christopher W. Lynn, Stephanie E. Palmer

overleaf.com/read/bkgkstkfvswr#db422d (manuscript in preparation)

Teaching

Grader, Biological Sciences Collegiate Division, University of Chicago Mar – Jun 2024

Grader for BIOS 20198: Biodiversity

Teaching Assistant, Biological Sciences Collegiate Division, UChicago Jan – Mar 2024

Lecture and laboratory teaching assistant for BIOS 10140: Inquiry-based Exploration of Biology

Conferences, Presentations & Workshops

APS Global Physics Summit, Anaheim, CA March 2025

Inferring minimax entropy models from neural responses to natural movies ([Talk](#))

NITMB Emerging Directions Workshop, Chicago, IL

February 2025

Leadership & Outreach

UChicago Society of Women in Physics — President

Sep 2023 – Jun 2024

UChicago SWiP — Treasurer, Webmaster, First-Year Representative

Sep 2020 – Jun 2023